From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Abstract

Intensive longitudinal research often seeks population-level inference on within-person dynamics without sacrificing idiographic estimation. We introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic systems that combines person-specific dynamic modeling with multivariate meta-analytic synthesis. In Stage 1, person-specific vector autoregressive (VAR) models are fit to obtain each individual's dynamic parameter vector and corresponding sampling covariance matrix. In Stage 2, those parameter vectors are synthesized using a multivariate random-effects meta-analytic SEM, allowing population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation while propagating first-stage uncertainty. In a Monte Carlo simulation, we compare MetaVAR with (a) a one-step Bayesian multilevel VAR benchmark implemented in `Mplus` under default or relatively uninformative priors and (b) an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark. Results indicate that MetaVAR's clearest advantage lies in recovery of between-person heterogeneity and calibration of interval estimates for heterogeneity parameters, whereas the one-step Bayesian benchmark remains competitive for population-average effects. An ecological momentary assessment illustration on negative and positive affect further shows that MetaVAR recovers interpretable average dynamics together with substantial heterogeneity in affective set points, inertia, cross-lagged coupling, and innovation structure.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, structural equation modeling

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Intensive longitudinal designs, such as ecological momentary assessment (EMA), experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols, have made it increasingly feasible to collect dense within-person data in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs have expanded opportunities to study psychological processes as dynamic systems, in which trajectories evolve over time within persons and individuals differ meaningfully in the parameters governing those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift reinforces a long-standing methodological point: inferences about within-person dynamics cannot be assumed from between-person variation, and questions about regulation, coupling, and stability require models that distinguish idiographic (person-specific) from nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018; Gates et al., 2023; Hamaker et al., 2005; Molenaar, 2004; Wang et al., 2026).

At the same time, applied researchers rarely want only a collection of person-specific estimates. They also want population-level summaries of average dynamics, between-person heterogeneity, and the ways dynamic features co-vary across people, as well as a basis for relating those person-specific dynamic signatures to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). Existing workflows, however, often force a trade-off between two unsatisfactory options. One-step hierarchical models with partial pooling—including multilevel and Bayesian multilevel formulations—can support population inference, but may become computationally demanding and shrink person-specific dynamics toward the population mean. At the other extreme, fully unpooled "fit-many-then-summarize" workflows are modular and scalable, but uncertainty-uncorrected versions often ignore first-stage uncertainty and the multivariate dependence among dynamic parameters, limiting valid inference on the dynamic system as a whole (Moeyaert et al., 2016; Van den Noortgate & Onghena, 2008).

This paper addresses this gap by introducing MetaVAR, a two-stage meta-analytic structural equation modeling framework (MASEM; Cheung, 2008, 2014; Cheung & Cheung, 2016) for multilevel dynamic models. In this manuscript, the Stage 1 idiographic model is a vector autoregressive model (VAR; Hamilton, 1994; Lütkepohl, 2005) fit separately to each individual, and Stage 2 synthesizes the resulting person-specific parameter vectors using a multivariate meta-analytic SEM with random effects. The key methodological contribution is not simply that MetaVAR is two-stage, but that it synthesizes the full person-specific dynamic parameter vector while carrying forward the full Stage 1 sampling covariance matrix. This enables population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation in the joint dynamic system rather than on isolated parameters one at a time. A further advantage of this formulation is that Stage 2 supports system-level hypotheses on the full dynamic parameter vector—including equality constraints, structured relations among parameters, and links to covariates or distal outcomes—rather than limiting inference to isolated coefficients considered one at a time.

The central question, then, is not only how to pool person-specific dynamic estimates, but how to do so in a way that preserves inferential validity while remaining practically feasible. Existing uncertainty-uncorrected two-stage workflows lose this validity by ignoring first-stage estimation error and the multivariate dependence among person-specific dynamic parameters, whereas one-step hierarchical models can be computationally demanding and partially pool person-specific estimates toward the population mean. MetaVAR is intended for settings in which researchers want to retain idiographic first-stage estimation while still obtaining valid multivariate population-level inference. In particular, MetaVAR is most useful when the scientific target includes not only population-average dynamics, but also between-person heterogeneity and cross-parameter covariation in the dynamic system. This practical niche is especially relevant in applications where person-specific models can be estimated independently and in parallel, making a two-stage workflow computationally attractive relative to joint hierarchical estimation. In what follows, we situate this contribution relative to existing one-step and two-stage approaches, evaluate it in a Monte Carlo simulation against a Bayesian multilevel VAR benchmark, operationalized here using the default dynamic structural equation modeling implementation in `Mplus`[1], alongside an uncertainty-uncorrected two-stage fit-many-then-summarize approach, and illustrate it using ecological momentary assessment data on negative and positive affect.

Two Dominant Strategies and a Persistent Gap

Two broad strategies dominate current practice for drawing population-level conclusions from intensive longitudinal multivariate time series. The first is hierarchical one-step (partially pooled) modeling—including multilevel and Bayesian multilevel formulations such as dynamic structural equation modeling (DSEM) and Bayesian multilevel VAR (mlVAR)—in which within-person dynamics and between-person variation are estimated simultaneously within a single hierarchical model (Asparouhov et al., 2018; Hamaker et al., 2018; Li et al., 2022). This approach is attractive in behavioral sciences because the length of the series per-person is often limited, making it useful to borrow strength across

[1] We treat this as a practical benchmark under a common default `Mplus` implementation rather than as a definitive evaluation of all Bayesian multilevel VAR specifications. In one-step Bayesian estimation, performance can depend materially on prior specification for random-effects covariance structures (Asparouhov & Muthén, 2024; Li et al., 2022; Schuurman et al., 2016; van Erp et al., 2018).

individuals while still allowing random effects in dynamic parameters (Li et al., 2022). At the same time, the estimation problem can become computationally demanding as the number of variables and random effects grows, and practical implementations often require simplifying assumptions about the random-effects structure or otherwise face feasibility and convergence constraints in higher-dimensional settings (Asparouhov et al., 2018; Epskamp et al., 2018; Li et al., 2022). Moreover, because hierarchical estimators partially pool information, person-specific dynamics are shrunk toward the population mean. Although this can improve stability and out-of-sample performance (Gelman & Hill, 2006), it also means that person-level estimates are biased toward the center and may be less suitable as direct measures of individual differences or for detecting unusual individuals (Efron & Morris, 1975).

At the other extreme are fully unpooled workflows that fit an idiographic model per person and then summarize the resulting coefficients informally, for example, by averaging. This "fit-many-then-summarize" strategy is modular and naturally parallelizable, but uncertainty-uncorrected summaries often treat person-specific estimates as if they were equally precise or measured without error, even though their precision can differ as a function of series length, missingness, and measurement quality (Hedges & Olkin, 1985; Pastor & Lazowski, 2017). In intensive time-series settings, dependence and model misspecification (e.g., unmodeled trends) can also affect precision and inference if not modeled (Baek et al., 2023; Shadish et al., 2008). A distinct limitation arises in multivariate dynamic systems: even when first-stage uncertainty is acknowledged, many multi-stage approaches do not retain the within-person sampling covariance matrix among dynamic parameters, making it difficult to synthesize the joint dynamic system at the second stage (Epskamp et al., 2018; Moeyaert et al., 2016). Valid two-stage inference therefore requires carrying forward each individual's first-stage sampling variance and, in multivariate settings, the full sampling covariance matrix rather than treating person-level estimates as error-free (Jackson et al., 2011; Riley, 2009).

Taken together, this basic trade-off between borrowing strength and idiographic flexibility is consistent with earlier comparisons of person-specific and multilevel autoregressive modeling, which showed that multilevel estimation can be advantageous when dynamic structure is relatively homogeneous or per-person series are short, whereas person-specific modeling becomes more attractive as dynamic heterogeneity increases and sufficient repeated observations are available (Liu, 2017). These trade-offs motivate a more principled two-stage alternative: estimate person-specific dynamic models first, then synthesize those results at the population level while explicitly accounting for first-stage uncertainty and dependence among estimated parameters. The gap is therefore not simply the absence of a two-stage workflow, but the absence of a two-stage workflow that supports valid multivariate population inference on the dynamic system as a whole.

Related Literatures: Idiographic Two-Stage Dynamic Modeling and Meta-Analytic SEM

MetaVAR brings together ideas from two related bodies of literature. The first is a growing literature on two-stage idiographic modeling for intensive longitudinal data, especially in dynamic network and related multivariate time-series traditions. Lee and Gates (2024) advocated a two-stage idiographic meta-analytic perspective for observational time series, emphasizing the importance of estimating within-person models idiographically and then using random-effects meta-analysis to characterize population distributions of within-person effects. Similarly, Zhang et al. (2025) showed, in the context of individual-level dynamic factor analysis (DFA), that ignoring first-stage estimation error can bias second-stage random-effects estimation. Taken together, this literature establishes both the scientific value of treating heterogeneity as a target of inference and the statistical importance of carrying first-stage uncertainty into second-stage synthesis.

The second body of literature is meta-analytic structural equation modeling (Cheung, 2008, 2014, 2018; Cheung & Cheung, 2016; Jak & Cheung, 2020). MASEM is attractive here because it can fit full path or factor models, account for sampling covariance among effect sizes, provide overall model fit, model heterogeneity in multivariate form, and accommodate dependent effect sizes in multilevel formulations (Cheung, 2008, 2014; Cheung & Cheung, 2016). SEM-based meta-analysis has also been used to compute multivariate effect sizes together with their sampling covariance matrices and to impose, test, or relax equality constraints as needed (Cheung, 2018). These are precisely the features needed for a Stage 2 model whose unit of inference is the full dynamic parameter system rather than one coefficient at a time. MetaVAR integrates these two literatures by embedding idiographic dynamic-model outputs within a multivariate meta-analytic SEM framework (Mehta & Neale, 2005).

More broadly, this framing aligns with a longer tradition of synthesizing person-specific or single-case studies. Earlier work discussed meta-analysis of applied time-series data; later work in single-case research developed hierarchical and multilevel methods for combining effect sizes across cases; and more recent work has extended those approaches to incorporate time, moderators, multiple outcomes, and heterogeneous within-case variance (Baek et al., 2023; Gingerich, 1984; Moeyaert et al., 2020; Ugille et al., 2012; Van den Noortgate & Onghena, 2003). MetaVAR extends this general logic to person-specific multilevel dynamic models by treating each individual's estimated dynamic signature as a multivariate effect-size vector to be synthesized jointly.

The move from path-specific synthesis to multivariate synthesis is not merely a technical generalization. In dynamic systems, substantive questions often concern configurations of parameters rather than isolated coefficients—for example, whether stronger autoregressive persistence co-occurs with stronger cross-lagged coupling, whether coupling is asymmetric across directions, or whether summary features of the system vary systematically across persons. Addressing such questions requires a second stage that treats the dynamic system itself as the unit of inference. MetaVAR is designed for that purpose.

MetaVAR: Dynamic MASEM for System-Level Pooling With Full Error Propagation

MetaVAR is a two-stage framework that treats multilevel dynamic modeling as a meta-analytic problem in which *persons play the role of studies.* In Stage 1, each individual $i = 1, \ldots, N$ is fit with an idiographic dynamic model, yielding a vector of estimated dynamic parameters and an accompanying sampling covariance matrix. In Stage 2, these person-level parameter vectors are synthesized using a multivariate random-effects meta-analytic model expressed as a structural equation model. The key feature of this design is that the full Stage 1 sampling covariance matrix is carried forward into Stage 2. As a result, MetaVAR supports population-level inference on the *joint* dynamic parameter system, including average dynamics, between-person heterogeneity, and cross-parameter covariation.

This perspective connects directly to multivariate random-effects meta-analysis (Berkey et al., 1998; van Houwelingen et al., 1993, 2002) and MASEM (Cheung, 2008, 2013, 2014; Cheung & Cheung, 2016). It is motivated by questions that arise frequently in intensive longitudinal research: What is the average within-person dynamic system? How much do people differ in that system? Which dynamic features tend to co-occur across people? More broadly, the same framework provides a basis for extensions involving person-level covariates or distal outcomes, although the present manuscript focuses on the core two-stage synthesis.[2]

Stage 1 Outputs as Multivariate Effect Sizes

Before defining the Stage 1 model formally, it is helpful to clarify why the VAR framework is useful in intensive longitudinal research. Like the random-intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015), the present centered VAR separates stable person-specific levels from the dynamics of deviations around those levels. In the intensive longitudinal setting, this makes it possible to study how a person's current state depends on prior deviations in the same variable and in other variables over time. The model therefore captures both within-variable carryover—often called autoregressive persistence or inertia—and cross-variable lagged relations, in which one process predicts subsequent change in another. In that sense, the lagged structure also bears a loose resemblance to actor–partner logic: autoregressive effects capture how a variable predicts itself over time, whereas cross-lagged effects capture how one process predicts subsequent change in another (Gistelinck & Loeys, 2018, 2020). As such, the person-specific VAR

[2] We use the label MetaVAR because the Stage 1 model in this manuscript is a VAR fit separately to each individual. More generally, however, the broader `metaDyn` framework requires only that each person contribute (a) a vector of estimated model features, $\hat{\boldsymbol{\theta}}_i$, and (b) an accompanying sampling covariance matrix, $\mathbf{V}_i^2$, so that first-stage uncertainty can be propagated into the population synthesis. In that sense, MetaVAR is the VAR-based instantiation of the broader `metaDyn` framework for dynamical systems.

provides a natural framework for studying short-term regulation, coupling, and spillover in multivariate psychological processes (Epskamp et al., 2018; Li et al., 2022).

More generally, let $\boldsymbol{\theta}_i \in \mathbb{R}^p$ denote the person-specific dynamic parameter vector for individual i, let $\hat{\boldsymbol{\theta}}_i$ denote its Stage 1 estimate, and let $\mathbf{V}_i^2 \in \mathbb{R}^{p \times p}$ denote the corresponding Stage 1 sampling covariance matrix. The contents of $\boldsymbol{\theta}_i$ are flexible: depending on the Stage 1 model, $\boldsymbol{\theta}_i$ may include autoregressive and cross-lagged coefficients, innovation variances and covariances, set point parameters, measurement-error variances, or other person-specific features of the dynamic system. The central requirement is that Stage 1 yield $\hat{\boldsymbol{\theta}}_i$ together with $\mathbf{V}_i^2$, or a good approximation to it, so that Stage 2 can propagate first-stage uncertainty rather than treat person-level estimates as error-free.

In this manuscript, Stage 1 fits a person-specific first-order VAR model. Let $\mathbf{s}_{i,t} \in \mathbb{R}^k$ denote the observed multivariate time series for person i at occasion $t = 1, \ldots, T$. The dynamic model follows a centered VAR specification (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015),

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i \left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i\right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}_i)\,, \tag{1}$$

where $\boldsymbol{\mu}_i$ is the person-specific set point, $\mathbf{A}_i$ collects autoregressive and cross-lagged effects governing the evolution of deviations from that set point, and $\boldsymbol{\Xi}_i$ is the innovation covariance matrix. Under stability or stationarity (e.g., all eigenvalues of $\mathbf{A}_i$ lie inside the unit circle) and mean-zero innovations, the process fluctuates around $\boldsymbol{\mu}_i$ such that $\mathbb{E}(\mathbf{s}_{i,t}) = \boldsymbol{\mu}_i$. For a bivariate process, the transition matrix may be written as

$$\mathbf{A}_i = \begin{pmatrix} a_{11,i} & a_{12,i} \\ a_{21,i} & a_{22,i} \end{pmatrix},$$

where the diagonal elements are autoregressive effects and the off-diagonal elements are cross-lagged effects. In the substantive language often used in affect-dynamics research, the autoregressive effects describe temporal persistence—often called *inertia*—in that they indicate how strongly deviations in a variable carry over to the next occasion. The cross-lagged effects describe lagged *coupling* between variables, indicating whether deviations in one process predict subsequent change in the other. We use these latter terms as descriptive labels for substantive interpretation, while retaining the standard VAR terminology in the formal model.

For the two-stage framework, Stage 1 contributes two outputs for each person: an estimated vector of dynamic features and a corresponding sampling covariance matrix that quantifies their estimation uncertainty. Stage 2 then treats those person-specific outputs as the objects to be synthesized at the population level.

For the present model, a convenient choice of person-specific parameter vector is

$$\boldsymbol{\theta}_i = \left(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)', \operatorname{vech}(\boldsymbol{\Xi}_i)'\right)', \tag{2}$$

where $(\cdot)'$ denotes matrix transpose, $\operatorname{vec}(\cdot)$ stacks a matrix by columns, and $\operatorname{vech}(\cdot)$ stacks the lower triangle of a symmetric matrix, including the diagonal. The associated Stage 1 sampling covariance matrix,

$$\mathbf{V}_i^2 = \widehat{\mathbb{V}}\left[\hat{\boldsymbol{\theta}}_i\right],$$

can be obtained from the observed information matrix, a sandwich estimator, or a bootstrap, and is then passed to Stage 2. If some components are fixed or omitted in a given application, they are simply excluded from $\boldsymbol{\theta}_i$, and the dimension p is adjusted accordingly.

Stage 2 as Multivariate Mixed-Effects Meta-Analysis in SEM Form

At Stage 2, the objects being modeled are no longer the raw repeated measures, but the person-specific parameter estimates obtained from Stage 1. The SEM formulation is therefore defined on the person-specific dynamic parameter system itself. This allows the second stage to model average dynamic features across persons, between-person heterogeneity in those features, and structured relations among them—for example, whether stronger inertia tends to co-occur with stronger cross-lagged coupling, or whether person-level covariates predict particular elements of the dynamic signature.

A convenient way to express this idea is through a measurement equation for the estimated person-specific parameters together with a structural model for the underlying person-specific parameter vector. Specifically, let

$$\hat{\boldsymbol{\theta}}_i = \boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \mathbf{V}_i^2\right), \tag{3}$$

so that the observed Stage 1 estimate is treated as a noisy measurement of the latent person-specific dynamic parameter vector, with known sampling covariance matrix $\mathbf{V}_i^2$. A general mixed-effects meta-analytic SEM for the latent parameter vector can then be written as (Cheung, 2008, 2013, 2014; Cheung & Cheung, 2016)

$$\boldsymbol{\theta}_i = \boldsymbol{\alpha}_i + \boldsymbol{\beta}\boldsymbol{\theta}_i + \boldsymbol{\Gamma}\mathbf{x}_i, \tag{4}$$

with person-specific latent parameter vectors

$$\boldsymbol{\alpha}_i = \boldsymbol{\alpha} + \boldsymbol{v}_i, \quad \boldsymbol{v}_i \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\tau}^2\right). \tag{5}$$

Here, $\boldsymbol{\alpha}$ is the grand-mean vector of the dynamic parameter system, $\boldsymbol{\tau}^2$ is the between-person heterogeneity covariance matrix, and $\mathbf{x}_i$ collects person-level covariates. The matrices $\boldsymbol{\beta}$ and $\boldsymbol{\Gamma}$ allow researchers to specify structured relations and meta-regressions defined *on the parameter system itself.* In the present manuscript, however, our primary focus is on the random-effects synthesis case, which sets $\boldsymbol{\beta} = \mathbf{0}$ and omits covariates.

Assuming that $\mathbf{x}_i$ is non-stochastic and that $\mathbf{I} - \boldsymbol{\beta}$ is nonsingular, Equation 4 can be written in reduced form (Bollen, 1989) as

$$\boldsymbol{\theta}_i = (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha}_i + \boldsymbol{\Gamma}\mathbf{x}_i) . \tag{6}$$

Combining this with Equation 3 yields the model-implied mean vector and covariance matrix of the observed Stage 1 estimates.[3]

$$\begin{aligned} \mathbb{E}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i) , \\ \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} \boldsymbol{\tau}^2 (\mathbf{I} - \boldsymbol{\beta})^{-1'} + \mathbf{V}_i^2. \end{aligned} \tag{7}$$

In the special case $\boldsymbol{\beta} = \mathbf{0}$, the model reduces to a multivariate mixed-effects meta-analysis, or to multivariate meta-regression when $\boldsymbol{\Gamma} \neq \mathbf{0}$, with

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i, \qquad \text{and} \qquad \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2.$$

When no covariates are included and $\boldsymbol{\beta} = \mathbf{0}$, the model further reduces to a multivariate random-effects meta-analysis with

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i\right] = \boldsymbol{\alpha}, \qquad \text{and} \qquad \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i\right] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2.$$

In this way, observed dispersion in $\hat{\boldsymbol{\theta}}_i$ is decomposed into (a) within-person estimation uncertainty carried by $\mathbf{V}_i^2$ and (b) genuine between-person heterogeneity carried by $\boldsymbol{\tau}^2$ (Pastor & Lazowski, 2017).

This system-level formulation is the key departure from edge-by-edge pooling. MetaVAR treats the full parameter vector as the unit of synthesis, thereby retaining the within-person sampling covariance

[3] In SEM software that supports *definition variables*, elements of the known, person-specific sampling covariance matrix $\mathbf{V}_i^2$ can be supplied as case-level data so that the model-implied covariance is recomputed for each individual (Lai & Hsiao, 2022; Mehta & Neale, 2005). This capability is central in meta-analytic applications because $\mathbf{V}_i^2$ is treated as a known input, whereas common mixed-effects implementations (e.g., `lmer`) do not directly support fixing a case-specific sampling covariance matrix in the likelihood (Declercq et al., 2022). In `OpenMx`, this is implemented by evaluating each individual's multivariate normal log-likelihood using that individual's model-implied mean vector and covariance matrix, which may depend on definition variables (FIML; Arbuckle, 1996; Enders, 2010; Neale et al., 2015). The `metaSEM` package leverages this capability via `OpenMx` (Cheung, 2015); the `metaDyn` implementation of MetaVAR uses the same strategy to incorporate $\mathbf{V}_i^2$ and propagate Stage 1 uncertainty through Stage 2.

among dynamic parameters and estimating a coherent between-person covariance structure for the dynamic system.

To describe heterogeneity, we report the conventional I^2 (Higgins & Thompson, 2002) separately for each dynamic parameter. For parameter p,

$$I_p^2 = \frac{\tau_p^2}{\tau_p^2 + \tilde{v}_p}, \tag{8}$$

where τ_p^2 is the p^{th} diagonal element of the between-person covariance matrix $\boldsymbol{\tau}^2$, and $\tilde{v}_p$ is the typical Stage 1 sampling variance for that parameter,

$$\tilde{v}_p = \frac{(N-1)\sum_{i=1}^{N} w_{i,p}}{\left(\sum_{i=1}^{N} w_{i,p}\right)^2 - \sum_{i=1}^{N} w_{i,p}^2}, \qquad \text{and} \qquad w_{i,p} = \frac{1}{v_{i,p}^2}, \tag{9}$$

with $v_{i,p}^2$ denoting the p^{th} diagonal element of $\mathbf{V}_i^2$ for individual i. Thus, I_p^2 gives the proportion of total variation in the estimated person-specific value of parameter p that is attributable to true between-person heterogeneity rather than Stage 1 estimation error. Values near zero indicate that most of the observed dispersion reflects sampling uncertainty, whereas values near one indicate substantial true heterogeneity.

Model-Level Hypotheses and Constraints

Because Stage 2 is an SEM defined on the full parameter vector, MetaVAR is naturally suited to *model-level* hypotheses, that is, hypotheses that jointly constrain or relate multiple dynamic parameters rather than treating each coefficient as an isolated edge. This matters because many substantive predictions about regulation are multivariate by construction. For example, in an affective system one might ask whether inertia is comparable across negative affect and positive affect, whether cross-lagged coupling is asymmetric, whether stronger inertia tends to co-occur with stronger cross-affect coupling across persons, or whether particular elements of the dynamic signature predict later outcomes such as symptom severity or treatment response.

In MetaVAR, such questions can be expressed through standard SEM machinery applied to $\boldsymbol{\alpha}$, $\boldsymbol{\tau}^2$, $\boldsymbol{\beta}$, and $\boldsymbol{\Gamma}$. Equality constraints can be used to test whether average inertia parameters are comparable across constructs; structured relations in $\boldsymbol{\beta}$ can be used to model interrelations among components of the dynamic signature itself; and covariates or distal outcomes can be linked to selected elements of the parameter system through $\boldsymbol{\Gamma}$. For instance, one could test whether individuals with stronger negative-affect inertia also tend to show more negative NA→PA coupling, or whether a distal outcome is predicted by a composite of set point, inertia, and innovation parameters rather than by any single coefficient alone.

This has three advantages. First, inference remains system-aware: constraints and regressions are evaluated while retaining both the within-person sampling covariance $\mathbf{V}_i^2$ and the between-person heterogeneity structure $\boldsymbol{\tau}^2$. Second, hypotheses may be formulated not only for individual coefficients but also for functions of the dynamic system, such as coupling contrasts, symmetry constraints, or derived summaries of stability. Third, competing theory-level models can be compared using familiar SEM tools, including likelihood-based comparisons for nested models and information criteria for non-nested models.

In the present manuscript, however, our focus is on the core two-stage synthesis: person-specific dynamic parameters are estimated idiographically and then synthesized using multivariate random-effects meta-analysis, cast as meta-analytic SEM, with full propagation of each individual's Stage 1 sampling covariance matrix. Extensions involving covariates, distal outcomes, or richer second-stage structures are possible within the same template, but we treat them as future directions rather than as central demonstrations here.

Evaluation and Empirical Illustration

The remainder of the paper evaluates the operating characteristics of MetaVAR and demonstrates its applied utility. We first conduct a Monte Carlo simulation study that varies sample size and per-person series length and focuses on recovery of population-average dynamics, between-person heterogeneity, and uncertainty quantification when Stage 1 sampling covariance matrices are propagated into Stage 2. We then illustrate the end-to-end workflow using intensive longitudinal affect data. We conclude with a general discussion of the method's practical implications, limitations, and extensions.

Monte Carlo Simulation Study

We conducted a Monte Carlo simulation study to evaluate the operating characteristics of MetaVAR under a controlled bivariate affect-dynamics data-generating process (DGP) and to compare it with two practically important alternatives representing the main inferential strategies discussed in the Introduction: a one-step hierarchical estimator, implemented here as a Bayesian multilevel VAR benchmark, and an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark. The simulation examined recovery of (a) population-average dynamic parameters, (b) between-person heterogeneity in those parameters, and (c) uncertainty quantification across combinations of sample size (N) and per-person time-series length (T).

Because MetaVAR is specifically designed to propagate Stage 1 sampling covariance into a multivariate second-stage synthesis, we expected its clearest advantages to emerge for recovery of between-person heterogeneity and for interval calibration for those heterogeneity parameters. At the same time, because the DGP and the Stage 1 estimation model were matched, the simulation represents a favorable, correctly specified setting for all methods.

Data-Generating Process

Data were generated using the `SimSSMVARIVary` function from the `simStateSpace` package (v1.2.15; Pesigan et al., 2025). The data-generating model was the same centered first-order VAR model used in Stage 1 (Equation 1) for person $i = 1, \ldots, N$ and occasion $t = 1, \ldots, T$.

The person-specific set points were generated as

$$\boldsymbol{\mu}_i \sim \mathcal{N}\left(\begin{pmatrix} 2.87 \\ 2.04 \end{pmatrix}, \begin{pmatrix} 1.20 & 0.46 \\ 0.46 & 1.10 \end{pmatrix}\right), \tag{10}$$

and the person-specific transition matrices were generated by drawing

$$\operatorname{vec}(\mathbf{A}_i) \sim \mathcal{N}\left(\begin{pmatrix} 0.280 \\ -0.035 \\ -0.048 \\ 0.260 \end{pmatrix}, \begin{pmatrix} 0.017 & 0.005 & 0.001 & 0.008 \\ 0.005 & 0.008 & 0.001 & 0.006 \\ 0.001 & 0.001 & 0.001 & 0.002 \\ 0.008 & 0.006 & 0.002 & 0.026 \end{pmatrix}\right). \tag{11}$$

Here, $\operatorname{vec}(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix into a single column vector. For a 2×2 matrix $\mathbf{A}_i = \left(a_{jk,i}\right)$, this implies $\operatorname{vec}(\mathbf{A}_i) = (a_{11,i},\, a_{21,i},\, a_{12,i},\, a_{22,i})'$.

Because the simulation allowed between-person heterogeneity only in the set points and transition parameters, the Stage 2 fixed-effect target can be written as

$$\boldsymbol{\alpha} = \left(\mathbb{E}[\boldsymbol{\mu}_i]', \mathbb{E}[\operatorname{vec}(\mathbf{A}_i)]'\right)'. \tag{12}$$

Likewise, the Stage 2 random-effects target is the between-person covariance matrix of $(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)')'$. Under the present DGP, this matrix is block diagonal because the set points and transition parameters were generated independently:

$$\boldsymbol{\tau}^2 = \begin{pmatrix} \boldsymbol{\Sigma}_\mu & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_A \end{pmatrix}. \tag{13}$$

This notation makes explicit that the fixed-effect targets correspond to $\boldsymbol{\alpha}$ and the heterogeneity targets correspond to $\boldsymbol{\tau}^2$.

To ensure a stationary VAR process for each individual, generated transition matrices were retained only if all eigenvalues of $\mathbf{A}_i$ had moduli less than 1 (i.e., spectral radius < 1). In practice, this was implemented via rejection sampling using `simStateSpace`'s `SimBetaN` helper: candidate draws of $\mathrm{vec}(\mathbf{A}_i)$ from the target multivariate normal distribution were repeatedly proposed until the corresponding $\mathbf{A}_i$ satisfied the stability constraint. Thus, the simulation targets a setting in which person-specific dynamics are heterogeneous but dynamically stable.

The innovation covariance matrix was held constant across persons and occasions,

$$\boldsymbol{\Xi} = \begin{pmatrix} 1.30 & 0.57 \\ 0.57 & 1.56 \end{pmatrix}, \tag{14}$$

so that the simulation isolates method performance for recovering heterogeneity in the set points and lagged dynamics. Fixing $\boldsymbol{\Xi}$ also simplifies interpretation because any between-person variation in estimated innovation parameters then reflects finite-sample estimation error rather than true heterogeneity in the DGP.

The parameter values were informed by prior empirical work on affective dynamics (Bringmann et al., 2013), but were not copied exactly. In particular, we parameterized the person-specific mean (set point) $\boldsymbol{\mu}_i$ rather than the intercept vector. This is substantively convenient because $\boldsymbol{\mu}_i$ directly represents each person's equilibrium level in the centered VAR parameterization, whereas the intercept in the uncentered form is a transformed quantity. The average transition matrix implies moderate positive autoregressive persistence for both variables and small negative cross-lagged effects, representing a stable bivariate regulatory system with weak reciprocal inhibitory coupling on average.

Design Factors

We primarily used a 3×3 factorial design crossing three sample sizes and three numbers of measurement occasions per individual:

$$N \in \{50, 100, 200\}, \qquad T \in \{50, 100, 200\}. \tag{15}$$

This yielded nine balanced conditions (Cases 1–9), where all individuals in a given replication shared the same series length T.

To induce heterogeneity in first-stage precision—and thus variability in the person-specific sampling covariance matrices used in Stage 2—we also included three unbalanced-T conditions (Cases 10–12). In these conditions, the sample size was fixed at $N \in \{50, 100, 200\}$ for Cases 10, 11, and 12, respectively, but the number of measurement occasions varied across individuals within the same replication. Specifically, for each individual i, the realized series length was $T_i \in \{50, 100, 200\}$, so that some individuals contributed shorter series and others longer series, producing heterogeneous $\mathbf{V}_i^2$ across persons. Analyses used each individual's realized T_i.

For each condition, we generated $R = 1{,}000$ Monte Carlo replications. In each replication, person-specific time series were generated under the DGP above, analyzed by each competing method, and summarized using the performance metrics described below.

Methods Compared

We compared three approaches representing the main inferential strategies discussed in the Introduction: (a) MetaVAR, a two-stage method with explicit propagation of first-stage uncertainty; (b) a one-step hierarchical benchmark, implemented here as Bayesian multilevel VAR (BMLVAR); and (c) an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark, which summarizes person-specific estimates at Stage 2 without propagating their sampling covariance matrices. Across all methods, the inferential targets were the same: population-average dynamic parameters (fixed effects) and between-person heterogeneity in those parameters (random effects), as defined by the simulation DGP.

To facilitate comparability, all methods were fit to the same simulated datasets under each (N, T) condition, and summaries were mapped to a common parameterization whenever possible (i.e., set points, VAR transition parameters, and between-person variability in those parameters). The comparison is therefore intended as a practical comparison of estimation strategies under a common target, not as a referendum on Bayesian versus frequentist inference.

MetaVAR

MetaVAR was implemented as a two-stage procedure using `fitVARMxID` (v1.0.2) for Stage 1 and `metaDyn` (v1.0.0) for Stage 2; both packages relied on `OpenMx` (v2.22.10; Hunter, 2017; Neale et al., 2015). In Stage 1, a person-specific VAR(1) model was fit separately for each individual to obtain (a) a vector of estimated dynamic parameters and (b) the corresponding estimated sampling covariance matrix. These outputs were then passed to Stage 2, where the person-level parameter vectors were synthesized using a multivariate random-effects meta-analytic model implemented in SEM form. Stage 2 treats the person-specific sampling covariance matrices as known inputs, thereby propagating first-stage estimation uncertainty into the population-level likelihood.

A key inferential feature of MetaVAR is that it operates on the full parameter vector for each person rather than pooling parameters one at a time. Consequently, MetaVAR carries forward the within-person sampling covariance among estimated dynamic parameters into Stage 2 and estimates a coherent between-person random-effects covariance structure across the parameter system. In the simulation, we evaluated MetaVAR using model-based normal-theory confidence intervals at Stage 2.

BMLVAR

BMLVAR was fit in `Mplus` (v9; Muthén & Muthén, 2017) using DSEM with two MCMC chains and 40,000 fixed iterations. We treat this analysis as a practical one-step benchmark under a common `Mplus` DSEM implementation with default or relatively uninformative priors, rather than as a definitive evaluation of Bayesian multilevel VAR more generally. This distinction matters because recovery of multilevel variance–covariance components can depend materially on prior specification, especially when heterogeneity parameters are small, weakly identified, or near the boundary (Asparouhov & Muthén, 2024; Li et al., 2022; Schuurman et al., 2016; van Erp et al., 2018).

Following `Mplus` guidance, Bayesian estimation was monitored using `TECH8` potential scale reduction (PSR), posterior effective sample size (ESS), and representative trace plots. In a representative simulation replication reported in the supplement, the maximum PSR reached 1.000 by the end of the run, and the 10 lowest ESS values ranged from 2,009 to 11,605. The lowest ESS values were concentrated in between-person random-effects variances and covariances for the lagged effects, indicating that posterior mixing was more demanding for heterogeneity parameters than for the fixed effects.

Because posterior credible intervals and frequentist confidence intervals are not identical in interpretation, coverage comparisons should be understood as operating-characteristic summaries under this particular prior and MCMC configuration.

Uncertainty-Uncorrected Two-Stage Benchmark

As a baseline two-stage approach, person-specific VAR(1) models were fit separately and then summarized across individuals without propagating first-stage estimation uncertainty in the second stage. Specifically, the second-stage analysis treated the Stage 1 point estimates as error-free and fit only the mean vector and covariance structure of the person-specific estimates using `OpenMx` (v2.22.10; Neale et al., 2015).

This benchmark reflects a common practical workflow in idiographic time-series research: fit models person-by-person, extract parameters of substantive interest, and then summarize those parameters across individuals. This general workflow is broadly consistent with common practice in person-specific network and multilevel VAR research, where individual models are often estimated separately and then compared or summarized across persons (Epskamp et al., 2018; Liu, 2017). This benchmark is included as a transparent and practically common baseline two-stage workflow, not as an exhaustive representation of all possible two-stage alternatives. Its advantages are modularity, transparency, and computational simplicity. However, because the second-stage summaries do not account for the within-person sampling covariance of the Stage 1 estimates, the estimated between-person covariance of the Stage 1 estimates reflects both true heterogeneity and first-stage sampling variability.

Comparative expectations. Relative to the uncertainty-uncorrected two-stage benchmark, MetaVAR was expected to improve population-level inference by carrying forward person-specific sampling uncertainty into the second stage. Relative to BMLVAR, MetaVAR trades one-step joint estimation for a modular two-stage workflow that preserves idiographic estimation while supporting multivariate population synthesis. The simulation therefore evaluates not only point-estimate recovery, but also the inferential consequences of these distinct strategies.

Performance Metrics

We evaluated performance separately for (a) population-average (fixed-effect) parameters and (b) between-person heterogeneity (random-effect) parameters.

Point-Estimate Performance

For each target parameter θ, we computed Monte Carlo relative bias and root mean square error (RMSE) across replications:

$$\begin{aligned} \text{Relative Bias}(\hat{\theta}) &= \frac{\bar{\hat{\theta}} - \theta}{\theta}, \\ \text{RMSE}(\hat{\theta}) &= \sqrt{R^{-1} \sum_{r=1}^{R} \left(\hat{\theta}_r - \theta\right)^2}, \\ \bar{\hat{\theta}} &= R^{-1} \sum_{r=1}^{R} \hat{\theta}_r. \end{aligned} \tag{16}$$

When summarizing the magnitude of relative bias without regard to direction, we used the absolute value of the relative bias: $\left|\text{Relative Bias}(\hat{\theta})\right|$.

Interval Performance (Empirical Coverage)

We evaluated uncertainty quantification using empirical coverage of nominal 95% interval estimates. For a given target parameter θ, empirical coverage was defined as

$$\widehat{\text{Coverage}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\theta \in \mathcal{I}_{r,.95}), \tag{17}$$

where $\mathcal{I}_{r,.95}$ denotes the nominal 95% interval estimate obtained in replication r.

For MetaVAR and the uncertainty-uncorrected two-stage benchmark, $\mathcal{I}_{r,.95}$ was the nominal 95% confidence interval. For BMLVAR, it was the nominal 95% posterior credible interval. Although these interval types are not identical in interpretation, the empirical inclusion rate of nominal 95% intervals provides a practically useful basis for comparing uncertainty quantification across methods.

Coverage values near .95 indicate well-calibrated interval estimation. As a secondary benchmark, Bradley's liberal robustness criterion implies that, for nominal 95% intervals, empirical coverage between approximately .92 and .97 may be regarded as acceptable (Bradley, 1978). We therefore refer to this metric generically as empirical coverage of nominal 95% intervals.

Hypothesis-Testing Performance (Empirical Power)

Because the present DGP did not include exact-zero target parameters, we evaluated inferential sensitivity using empirical power for parameters with nonzero true values rather than empirical Type I error. For a given target parameter θ and nominal level α,

$$\widehat{\text{Power}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\text{reject } H_0 : \theta = 0 \text{ in replication } r)\,. \tag{18}$$

For MetaVAR and the uncertainty-uncorrected two-stage benchmark, rejection was based on whether the nominal 95% confidence interval excluded zero. For BMLVAR, we report the empirical detection rate based on whether the nominal 95% posterior credible interval excluded zero. Because several true effects in the DGP were small in magnitude—especially the cross-lagged effects and some heterogeneity components—empirical power is interpreted jointly with bias, RMSE, and coverage.

Monte Carlo Simulation Results

Simulation performance was evaluated using relative bias, RMSE, empirical coverage of nominal 95% intervals, and empirical power. The clearest summary is that the comparative pattern depended on the inferential target. Aggregated across all parameters and design cells, MetaVAR showed the best overall balance of point-estimate performance and interval calibration in the present implementation, with the smallest mean absolute relative bias ($|\text{Relative bias}| = 0.186$), the lowest mean RMSE (0.043), and the mean coverage closest to nominal (0.926). BMLVAR was intermediate ($|\text{Relative bias}| = 1.015$, RMSE $= 0.047$, coverage $= 0.760$), whereas the uncertainty-uncorrected two-stage benchmark produced the highest mean power (0.810) but also the largest mean absolute relative bias (1.540), a higher mean RMSE (0.046), and the poorest coverage (0.636). This aggregate summary should be interpreted cautiously, however, because it combines fixed and random effects and because the BMLVAR benchmark reflects a specific `Mplus` implementation with default or relatively uninformative priors. The more informative comparison is therefore the contrast by inferential target.

Population-Average Parameters

For population-average parameters, all three methods showed small average bias, and BMLVAR performed best overall in interval calibration. Across fixed effects, mean absolute relative bias was small for all methods (0.005 for BMLVAR, 0.024 for MetaVAR, and 0.039 for the uncertainty-uncorrected two-stage benchmark), with BMLVAR yielding the best mean coverage (0.962), followed by MetaVAR (0.911) and the uncertainty-uncorrected benchmark (0.860). Mean power for fixed effects was high for all three methods (approximately 0.92). Thus, for recovery of population-average dynamics alone, the one-step Bayesian benchmark was highly competitive and, in terms of fixed-effect coverage, performed best in the present simulation.

Heterogeneity Parameters

The clearest advantage of MetaVAR emerged for the heterogeneity parameters. Across random-effects variances and covariances, MetaVAR yielded substantially smaller mean absolute relative bias (0.260) and lower mean RMSE (0.039) than either BMLVAR (1.482 and 0.047, respectively) or the uncertainty-uncorrected two-stage benchmark (2.233 and 0.044, respectively). MetaVAR also showed markedly better interval calibration for heterogeneity parameters, with mean coverage of 0.932, compared with 0.666 for BMLVAR and 0.533 for the uncertainty-uncorrected benchmark. This pattern is central to the rationale for MetaVAR: once the scientific target includes between-person heterogeneity in the dynamic system, carrying forward Stage 1 sampling covariance materially improves second-stage inference relative to treating person-specific estimates as error-free.

Because several heterogeneity parameters were close to zero, relative bias was naturally amplified for some components. For that reason, RMSE and empirical coverage are especially informative for interpreting performance on those parameters.

Figure 1 provides an overall summary of simulation performance by inferential target, separating population-average parameters (fixed effects) from between-person heterogeneity parameters (random effects). Figure 2 then shows parameter-level empirical coverage of nominal 95% intervals across methods and design conditions. The remaining parameter-level heatmaps for absolute relative bias, RMSE, and power are provided in the online supplement. The overview in Figure 1 makes clear that the comparative pattern depended on the inferential target. For population-average parameters, all three methods performed similarly in point estimation, and BMLVAR showed the best overall interval calibration. For between-person heterogeneity parameters, however, MetaVAR showed the strongest overall performance, with lower absolute relative bias and RMSE and markedly better coverage than either BMLVAR or the uncertainty-uncorrected benchmark. Figure 2 further shows that the clearest differences were concentrated in interval calibration: undercoverage was most severe for the uncertainty-uncorrected benchmark, BMLVAR was generally intermediate for heterogeneity parameters, and MetaVAR remained closest to nominal coverage across most random-effects targets and design cells.

Sample size primarily affected RMSE and power. For MetaVAR, mean RMSE decreased from 0.057 at $N = 50$ to 0.030 at $N = 200$, while mean power increased from 0.618 to 0.809. BMLVAR showed a similar pattern, with mean RMSE decreasing from 0.066 to 0.031 and mean power increasing from 0.678 to 0.842 as sample size increased from $N = 50$ to $N = 200$. The uncertainty-uncorrected two-stage benchmark also showed lower RMSE and higher power with larger N (RMSE: 0.061 to 0.034; power: 0.738 to 0.875), but its coverage deteriorated as sample size increased (0.741 at $N = 50$ vs. 0.529 at $N = 200$). For MetaVAR, average coverage remained relatively close to nominal across sample sizes (0.933, 0.929, and 0.914 for $N = 50$, 100, and 200, respectively), whereas BMLVAR remained consistently below nominal across all N conditions (0.783, 0.756, and 0.740).

The number of measurement occasions had an even clearer impact on performance, especially for MetaVAR. For MetaVAR, mean absolute relative bias decreased from 0.475 at $T = 50$ to 0.040 at $T = 200$, while mean power increased from 0.651 to 0.782 and mean coverage improved from 0.903 to 0.935. The unbalanced-T condition performed similarly to the balanced $T = 100$ condition, with mean absolute relative bias of 0.099, RMSE of 0.042, coverage of 0.933, and power of 0.717. BMLVAR also improved as the number of occasions increased, although more modestly: mean absolute relative bias decreased from 1.207 at $T = 50$ to 0.894 at $T = 200$, RMSE decreased from 0.049 to 0.045, and power increased from 0.698 to 0.819, but coverage remained low and essentially unchanged (between 0.754 and 0.770). The uncertainty-uncorrected two-stage benchmark showed substantial improvement as T increased, with mean absolute relative bias decreasing from 2.758 at $T = 50$ to 0.601 at $T = 200$ and coverage increasing from 0.518 to 0.764, but even in the most favorable $T = 200$ condition its average coverage remained well below nominal.

This distinction between fixed effects and heterogeneity parameters was especially pronounced under short series. For MetaVAR, average fixed-effect coverage was clearly below nominal when $T = 50$, whereas heterogeneity-parameter coverage remained much closer to the acceptable range. By contrast, the uncertainty-uncorrected two-stage benchmark showed severe undercoverage for heterogeneity parameters throughout, especially when T was short, consistent with the expectation that ignoring Stage 1 estimation error inflates apparent between-person variability. BMLVAR, although strong for fixed effects, also showed persistent undercoverage for heterogeneity parameters, suggesting that the present one-step benchmark did not fully resolve recovery of the population covariance structure under this design.

Computational speed under parallel execution also favored the two-stage workflow. In a benchmark dataset with $N = 200$ individuals and $T = 200$ measurement occasions per individual, mean runtimes were approximately 14.72 seconds for the uncertainty-uncorrected two-stage benchmark, 46.64 seconds for MetaVAR, and 363.42 seconds for BMLVAR. Thus, although MetaVAR incurs additional computation relative to a simple fit-many-then-summarize workflow, it remained substantially faster than joint Bayesian estimation in this setting. This advantage is especially relevant because Stage 1 idiographic estimation is embarrassingly parallel: person-specific models can be fit independently across multiple CPU cores or computing nodes.

Taken together, these results show that MetaVAR's main advantage lies not in maximizing empirical power, but in recovering between-person heterogeneity with better-calibrated interval estimates. BMLVAR remained highly competitive for population-average dynamics, especially fixed-effect coverage, whereas the uncertainty-uncorrected benchmark often achieved higher apparent power at the cost of substantial bias and undercoverage. MetaVAR is therefore most attractive when the scientific target includes heterogeneity in dynamic parameters and principled uncertainty quantification is essential.

Monte Carlo Simulation Discussion

The simulation supports the central rationale for MetaVAR as a two-stage framework for system-level inference that propagates Stage 1 uncertainty. Its clearest advantage emerged for between-person heterogeneity in the dynamic parameter system. Across random-effects variances and covariances, MetaVAR generally outperformed both the `Mplus`-based BMLVAR benchmark and the uncertainty-uncorrected two-stage benchmark in bias, RMSE, and coverage, indicating that carrying the full person-specific sampling covariance matrix into Stage 2 improves inference for heterogeneity parameters. By contrast, for population-average parameters, BMLVAR was highly competitive and in the present design provided the best overall fixed-effect coverage. The results therefore do not suggest that MetaVAR uniformly dominates one-step hierarchical estimation; rather, they indicate that MetaVAR is especially useful when the scientific target includes the between-person covariance structure of the dynamic system, not only its average dynamics.

These conclusions should be interpreted in light of the present design, which was favorable and correctly specified for all methods, and in light of the specific Bayesian implementation used for the one-step benchmark. In particular, the weaker recovery of heterogeneity parameters by BMLVAR should not be read as a definitive statement about Bayesian multilevel VAR under alternative prior specifications, additional chains, longer runs, or different software implementations. Rather, the present comparison situates MetaVAR as a principled two-stage alternative when researchers want to preserve idiographic estimation, exploit parallel computation, and obtain better-calibrated multivariate population inference on heterogeneity.

Empirical Illustration: Meta-Analysis of Affect Dynamics

We illustrate the proposed MetaVAR workflow using intensive longitudinal data from the Affective Dynamics and Individual Differences study (ADID; Emotions & Dynamic Systems Laboratory, 2010), which was designed to examine how emotion dynamics unfold in everyday life and how these dynamics differ across individuals. The ADID EMA protocol sampled participants five times per day for approximately 30 days (four daytime prompts plus an evening report), yielding up to 150 measurement occasions per participant in the raw data. ADID has contributed to multiple substantive and methodological studies, including work on regime-switching dynamics, measurement/trait–state structure, network and VAR-based modeling, outlier-robust state-space modeling, and missing-data methods (Chow & Zhang, 2013; Gates et al., 2023; Guo et al., 2012; Hutton & Chow, 2014; Li et al., 2022, 2024; Oh et al., 2025; Ou et al., 2023; Park et al., 2020; You et al., 2019).

This setting closely matches the goal of MetaVAR: estimate person-specific dynamic signatures in a first stage and then synthesize them at the population level in a second stage while propagating each individual's Stage 1 sampling uncertainty. In the present illustration, we focus on within-person dynamics between negative affect (NA) and positive affect (PA). Specifically, we meta-analyze the full person-specific parameter system consisting of (a) affective set points, (b) temporal dynamics encoded by the VAR(1) transition matrix—autoregressive (AR) inertia and cross-lagged coupling—and (c) the innovation covariance matrix, which captures within-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Substantively, NA and PA are useful targets because emotion-dynamics theories treat affect as a self-regulating system characterized by a typical level (often described as a set point), temporal persistence from one occasion to the next (often described as inertia), and possible cross-lagged coupling between affect dimensions that may reflect carryover or regulatory spillover (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). MetaVAR treats heterogeneity in these features—including heterogeneity in the innovation covariance structure—as a scientific target, providing population-average inference while quantifying between-person variation and co-variation across the dynamic system.

Measures

Momentary affect in ADID was assessed using adjective ratings drawn from established affect instruments, including items from the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988) and circumplex-content items used to broaden coverage of activation levels (Russell, 1980).

Participants rated how often they had experienced each affect descriptor since the previous assessment on an ordinal Likert-type scale, as implemented in the ADID daily diary surveys.

Following prior ADID applications that regularized the irregular EMA observations to equally spaced intervals for discrete-time modeling and applied within-person linear detrending, we constructed per-occasion NA and PA composites by averaging the available NA and PA items (Park et al., 2020; You et al., 2019). We then rounded the recorded date–time stamps to the nearest hour and inserted missing values for hours with no report to obtain an approximately hourly series for the present VAR analysis. These steps align the data with the discrete-time VAR framework used here, so the lagged coefficients are interpreted as approximately hourly dependencies in detrended affect rather than as features of the full irregularly spaced raw month-long trajectory.

Results

Stage 1 estimated person-specific NA–PA dynamics for each participant, and Stage 2 synthesized these individual dynamic signatures to obtain (a) population-average dynamics, (b) between-person heterogeneity in those dynamics, and (c) systematic co-variation among elements of the parameter system.

Stage 1: Person-Specific VAR

We fit the person-specific VAR(1) model in Equation 1 separately for each participant using the `fitVARMxID` package (v1.0.2), estimating all dynamic parameters as person-specific: set points, autoregressive coefficients, cross-lagged coefficients, and innovation covariance parameters. In the processed dataset, person-specific VAR models were fit for 217 individuals, and the Stage 1 estimation converged for all individuals in the empirical analysis.

Stage 2: Multivariate Random-Effects Meta-Analysis of the Dynamic Parameter System

Stage 2 synthesized the person-specific VAR(1) parameter systems using a multivariate random-effects meta-analysis with the `metaDyn` package (v1.0.1). The meta-analytic mean vector $\boldsymbol{\alpha}$ contained nine elements, ordered as the two set points ($\mu_{\mathrm{NA}}, \mu_{\mathrm{PA}}$), the four VAR transition coefficients ($a_{11}, a_{21}, a_{12}, a_{22}$), and the three unique elements of the innovation covariance matrix $\boldsymbol{\Xi}$ for NA and PA ($\xi_{11}, \xi_{21}, \xi_{22}$).

Population-average dynamics ($\boldsymbol{\alpha}$). The population-average set points indicated higher typical PA than NA: $\mu_{\mathrm{NA}} = 1.4402$ (SE = 0.0253), 95% CI [1.3906, 1.4897], and $\mu_{\mathrm{PA}} = 2.5817$ (SE = 0.0322), 95% CI [2.5187, 2.6448]. The average temporal dynamics showed positive inertia in both affect dimensions, with $a_{11} = 0.3571$ (SE = 0.0311), 95% CI [0.2961, 0.4181] for NA and $a_{22} = 0.3167$ (SE = 0.0313), 95% CI [0.2552, 0.3781] for PA.

Both cross-lagged effects were negative on average. The NA→PA effect was $a_{21} = -0.0728$ (SE = 0.0367), 95% CI [-0.1447, -0.0009], and the PA→NA effect was $a_{12} = -0.0593$ (SE = 0.0187), 95% CI [-0.0959, -0.0227]. Thus, higher NA at one occasion predicted lower subsequent PA on average, and higher PA predicted lower subsequent NA on average. Interpreted descriptively, these estimates are consistent with reciprocal inhibitory coupling in the short-term affective system at the hourly timescale used here.

Stage 2 also estimated the population-average innovation covariance matrix in the original metric, with $\xi_{11} = 0.0333$ (SE = 0.0019), 95% CI [0.0296, 0.0370], $\xi_{21} = -0.0073$ (SE = 0.0010), 95% CI [-0.0093, -0.0054], and $\xi_{22} = 0.0609$ (SE = 0.0029), 95% CI [0.0552, 0.0666]. Thus, after accounting for lagged dynamics, the innovation variance was larger for PA than for NA, and the innovation covariance was negative, indicating that within-occasion shocks to NA and PA tended to move in opposite directions on average.

Between-person heterogeneity and co-variation ($\boldsymbol{\tau}^2$). The estimated random-effects covariance matrix $\boldsymbol{\tau}^2$ indicated substantial heterogeneity across the full parameter system. The diagonal elements (between-person variances) were 0.1373 for μ_{NA} (SD $\approx$ 0.37), 0.2227 for μ_{PA} (SD $\approx$ 0.47), 0.1867 for a_{11} (SD $\approx$ 0.43), 0.2295 for a_{21} (SD $\approx$ 0.48), 0.0599 for a_{12} (SD $\approx$ 0.24), 0.1902 for a_{22} (SD $\approx$ 0.44), and, for the innovation parameters, 0.0007 for ξ_{11} (SD $\approx$ 0.027), 0.0001 for ξ_{21} (SD $\approx$ 0.010), and 0.0016 for ξ_{22} (SD $\approx$ 0.040).

Several, though not all, off-diagonal elements were statistically distinguishable from zero, indicating systematic co-variation among components of the dynamic signature across individuals. For example, individuals with higher NA set points tended to show greater NA inertia (cov = 0.0273, $p = .0212$). NA inertia a_{11} covaried negatively with the NA→PA coupling a_{21} (cov = -0.0684, $p < .001$), and PA inertia a_{22} covaried negatively with the PA→NA coupling a_{12} (cov = -0.0224, $p = .0130$). There was also evidence of co-variation between set points and innovation parameters, including μ_{NA} with ξ_{11} (cov = 0.0040, $p < .001$) and ξ_{22} (cov = -0.0022, $p = .0373$), and μ_{PA} with ξ_{11} (cov = -0.0022, $p = .0126$) and ξ_{22} (cov = -0.0055, $p < .001$). In addition, several covariances among the innovation parameters were statistically significant, including ξ_{11} with ξ_{21} (cov = -0.0002, $p < .001$), ξ_{11} with ξ_{22} (cov = 0.0004, $p < .001$), and ξ_{21} with ξ_{22} (cov = -0.0002, $p < .001$), consistent with meaningful between-person differences in the magnitude and co-fluctuation of within-occasion shocks.

Heterogeneity index (I^2). The I^2 indices were high for all nine elements of $\boldsymbol{\alpha}$, indicating that the observed dispersion in Stage 1 parameter estimates was dominated by true between-person heterogeneity rather than by within-person sampling variability. Under Equation 8, this means that the estimated between-person heterogeneity dominated the typical Stage 1 sampling variance for each parameter. Specifically, I^2 was 0.9990 for μ_{NA}, 0.9975 for μ_{PA}, 0.9636 for a_{11}, 0.9448 for a_{21}, 0.9555 for

a_{12}, 0.9514 for a_{22}, 0.9994 for ξ_{11}, 0.9944 for ξ_{21}, and 0.9954 for ξ_{22}. Taken together, the $\boldsymbol{\tau}^2$ and I^2 results support the motivating premise of MetaVAR: affective set points, lagged dynamics, and innovation structure all vary meaningfully across individuals, making multivariate synthesis of the full parameter system scientifically informative.

Empirical Example Discussion

This illustration shows how MetaVAR can be used to estimate idiographic affect-dynamics parameters from intensive longitudinal data and then synthesize those person-specific signatures into population-level inferences while quantifying heterogeneity. In ADID, the approach yielded interpretable population-average dynamics, strong evidence of between-person heterogeneity in both set points and lagged dependencies, and substantively meaningful co-variation among elements of the parameter system (Houben et al., 2015). In addition to the familiar temporal dependencies encoded in the VAR transition matrix, MetaVAR also supports population inference on the innovation structure, which captures within-occasion co-fluctuation in unexpected affective shifts after accounting for lagged dynamics.

Population-Average NA–PA Dynamics

At the population level, participants reported higher typical PA than NA, consistent with work showing that average affective tone in community samples tends to be more positive than negative (Houben et al., 2015; Watson et al., 1988). Both NA and PA showed positive inertia, with AR(NA) slightly larger than AR(PA), suggesting that momentary elevations in NA persisted somewhat more strongly than elevations in PA at the hourly scale used here (Houben et al., 2015; Kuppens, Oravecz, & Tuerlinckx, 2010). Cross-lagged coupling was negative in both directions: higher NA predicted lower subsequent PA, and higher PA predicted lower subsequent NA. Interpreted as a descriptive bivariate regulatory system, these results are broadly consistent with reciprocal inhibition or spillover processes discussed in affect-dynamics theory (Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). At the same time, these coefficients should be interpreted as average short-term dependencies at a specific time scale (hourly, after detrending), not as mechanistic causal effects (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015).

These findings are partly consistent with earlier ADID analyses, which likewise supported lagged affective dependencies but suggested a more asymmetric cross-lagged pattern. In particular, You et al. (2019), using a coarser equally spaced aggregation of the irregular EMA observations together with within-person linear detrending, suggested stronger evidence for a PA→NA lagged effect than for the reverse NA→PA effect. The present MetaVAR analysis instead suggests negative average cross-lagged effects in both directions. One plausible reason is timescale: the present study regularized the series to an approximately hourly grid, whereas the earlier state-space analysis used a coarser temporal aggregation.

Another difference is inferential target. Earlier ADID analyses were not designed to estimate a full random-effects covariance structure for the person-specific dynamic parameter system. MetaVAR therefore adds information those analyses could not directly target, including between-person heterogeneity in inertia and cross-lagged coupling as well as covariation among these dynamic features across individuals (Park et al., 2020; You et al., 2019).

The population-average innovation covariance matrix further indicated that, after accounting for lagged dynamics, within-occasion unexpected shifts in NA and PA tended to move in opposite directions on average. The larger innovation variance for PA than for NA suggests that short-term unexplained fluctuations were greater for PA in this dataset.

Heterogeneity as a Substantive Finding, Not a Nuisance

The random-effects variance components indicated substantial between-person variability across the dynamic signature, including set points, lagged dynamics, and innovation parameters. The corresponding I^2 values were uniformly high, implying that the observed dispersion in Stage 1 estimates overwhelmingly reflected true between-person differences rather than sampling variability (Higgins & Thompson, 2002). Methodologically, this pattern motivates multivariate synthesis rather than edge-by-edge pooling, because heterogeneity was present across parameters and the off-diagonal elements of $\boldsymbol{\tau}^2$ showed how deviations in set points, inertia, coupling, and innovation structure co-occurred across individuals. Because $I_p^2 = \tau_p^2/(\tau_p^2 + \tilde{v}_p)$ in the present framework, values near 1 arise when true between-person heterogeneity is large relative to the typical Stage 1 sampling variance. In that sense, the uniformly high I^2 values here indicate that the spread in person-specific affect-dynamics estimates was driven primarily by genuine differences across individuals rather than by estimation noise.

Several examples illustrate this point. Individuals with higher NA set points tended to show greater NA inertia, suggesting that typical affective level and short-term persistence may be linked within persons. Likewise, stronger NA inertia was associated with more negative NA→PA coupling, and stronger PA inertia was associated with more negative PA→NA coupling, indicating that temporal persistence and cross-affect coupling were not independent sources of heterogeneity. Co-variation among the innovation parameters further suggests meaningful between-person differences in the magnitude and structure of within-occasion affective shocks.

Practical and Methodological Considerations

Several features of the analysis highlight common trade-offs in applied dynamic modeling. First, we imposed regular spacing by rounding responses to the nearest hour and inserting missing values for unobserved hours, and we applied within-person linear detrending to better align the data with a stationary VAR(1) approximation. These choices improve feasibility and interpretability within the present discrete-time framework, but they also mean that the empirical target is a detrended, approximately hourly affective process rather than the full irregularly spaced raw EMA record, shifting interpretation toward shorter-term fluctuations rather than slower drift across the month-long protocol (Guo et al., 2012; Li et al., 2022). Second, NA and PA composites were constructed from ordinal item ratings and treated as approximately continuous. Future work could examine whether latent measurement models or item-level analyses alter the estimated dynamic signature or its heterogeneity (Park et al., 2020; Watson et al., 1988).

Implications and Future Directions

Overall, the illustration supports the value of treating affect dynamics as person-specific signatures that can be meaningfully compared and synthesized (Fisher et al., 2017; Houben et al., 2015). Several extensions follow naturally from the same Stage 2 SEM template but are not pursued here, including covariate moderation of the parameter system and links between dynamic signatures and distal outcomes under full Stage 1 error propagation. Additional directions include allowing nonstationarity (e.g., regime switching or time-varying parameters; Chow & Zhang, 2013; Hutton & Chow, 2014), extending Stage 1 to continuous-time formulations to better handle irregular sampling without discretization (Park et al., 2025), and targeting multivariate constraints or derived functionals—such as stability summaries, impulse-response metrics, or innovation-based summaries—as higher-level quantities for synthesis. A natural next step is to place person-level emotion-regulation markers and related covariates on the Stage 2 parameter system to explain why individuals differ in set points, inertia, coupling, and innovation structure.

General Discussion and Future Directions

The central aim of this paper was to develop a principled way to move from person-specific dynamic models to population-level inference without collapsing the distinction between within-person and between-person structure. MetaVAR addresses this aim by combining idiographic estimation in Stage 1 with multivariate meta-analytic structural equation modeling in Stage 2. In doing so, it brings together two lines of work that are usually kept separate: recent two-stage idiographic dynamic-modeling approaches that treat heterogeneity as a target of inference rather than a nuisance (Lee & Gates, 2024; Zhang et al., 2025), and the MASEM literature, which provides machinery for modeling multivariate effect sizes, sampling covariances, heterogeneity, and structured parameter relations within a single SEM framework (Cheung, 2008, 2014; Cheung & Cheung, 2016; Jak & Cheung, 2020). In this sense, MetaVAR is not simply a new way to average person-specific coefficients. It is a system-level synthesis framework in which persons play the role of studies and each person's dynamic signature is treated as a multivariate effect-size vector. More broadly, the present contribution complements earlier comparisons of person-specific and multilevel autoregressive modeling (Liu, 2017) by showing that the relevant methodological question is not only whether within-person dynamics should be estimated idiographically or hierarchically, but also how idiographic estimates should be synthesized once obtained if the goal is valid population-level inference on heterogeneity and cross-parameter co-variation.

This positioning also clarifies where MetaVAR fits in the broader methodological landscape. MetaVAR is not intended as a universal replacement for one-step hierarchical modeling. When the main scientific target is the population-average dynamic structure, or when partial pooling is itself substantively desirable, one-step multilevel models remain highly attractive. Consistent with that, the Bayesian multilevel VAR benchmark was highly competitive for population-average parameters in the present simulation. At the same time, that comparison should be interpreted as a practical benchmark based on a common `Mplus` implementation with default or relatively uninformative priors, not as a definitive evaluation of all Bayesian multilevel VAR specifications. In one-step Bayesian estimation, recovery of variance–covariance components can depend materially on prior specification for random-effects covariance structures, especially when those parameters are small or weakly identified (Asparouhov & Muthén, 2024; Li et al., 2022; Schuurman et al., 2016; van Erp et al., 2018). The present results therefore speak most directly to the practical trade-offs among these particular implementations rather than to a general Bayesian–frequentist contrast.

MetaVAR's clearest contribution is to valid two-stage multivariate inference when the scientific target includes between-person heterogeneity and cross-parameter co-variation in the dynamic system. By carrying forward each person's Stage 1 sampling covariance matrix $\mathbf{V}_i^2$ rather than treating person-specific estimates as error-free, MetaVAR separates first-stage estimation noise from genuine between-person differences and retains dependence among estimated dynamic parameters. This matters because, in multivariate meta-analysis more generally, ignoring within-study covariance can distort estimation and inference, particularly when the goal is to synthesize multiple correlated outcomes jointly (Berkey et al., 1998; Jackson et al., 2011; Riley, 2009; van Houwelingen et al., 2002). The simulation results were consistent with that logic. MetaVAR did not uniformly dominate the comparison methods across all targets, but it showed its clearest advantage for the recovery of between-person heterogeneity and for the calibration of interval estimates for heterogeneity parameters. That distinction is substantively important because many questions in intensive longitudinal research concern not only what dynamics look like on average, but also how and why people differ in those dynamics.

The empirical illustration reinforces this point. In the ADID data, MetaVAR recovered interpretable population-average affect dynamics while also revealing substantial heterogeneity in set points, autoregressive persistence, cross-lagged coupling, and innovation structure. More importantly, the off-diagonal elements of $\boldsymbol{\tau}^2$ showed that these features did not vary independently across people: dynamic characteristics such as inertia, coupling, and innovation structure co-varied in systematic ways. This extends earlier ADID analyses that were informative about average lagged dynamics and preprocessing choices, but were not designed to estimate a full random-effects covariance structure for the person-specific parameter system (Park et al., 2020; You et al., 2019). At the same time, the present empirical target should be interpreted carefully. Because the series were regularized to approximately hourly bins and detrended before estimation, the resulting parameters describe a short-term, detrended, approximately hourly affective process rather than the full irregularly spaced raw month-long EMA trajectory.

Several limitations qualify these conclusions. The simulation considered a bivariate data-generating process, fixed innovation covariance across persons, and correct model specification. The empirical illustration relied on observed composites, discretization of irregular measurements, and a stationary VAR(1) approximation after detrending. Accordingly, the present results speak to performance under a favorable simulation setting and to one practically useful but simplified empirical workflow. Future work should examine higher-dimensional systems, heterogeneity in innovation processes, missing data and irregular spacing, model misspecification, and Stage 1 models beyond observed-variable VARs, including latent-variable, state-space, continuous-time, and nonlinear formulations. It will also be important to study the behavior of MetaVAR when Stage 1 sampling covariance matrices are themselves approximate, such as when they are obtained via Taylor-series approximations, sandwich estimators, or bootstrap procedures.

Several extensions are especially promising. Because Stage 2 is formulated as an SEM on the parameter system itself, person-level covariates can be used to explain why some individuals show stronger inertia, weaker cross-regulation, or different innovation structures than others, and distal outcomes can be regressed on the dynamic signature under full propagation of Stage 1 uncertainty. In the affect-dynamics context, a natural next step is to examine whether person-level emotion-regulation markers or related individual differences account for heterogeneity in set points, inertia, coupling, and innovation parameters, and whether those dynamic signatures predict clinically or developmentally meaningful endpoints. More broadly, the same framework can be used to synthesize derived functionals rather than raw parameters, such as stability summaries, coupling contrasts, impulse-response quantities, or other theory-relevant summaries of a dynamic system.

Overall, MetaVAR contributes a principled and flexible framework for drawing population-level conclusions from person-specific dynamics. Its most distinctive value is not that it replaces one-step hierarchical estimation in all settings, but that it provides valid multivariate population inference on heterogeneity and cross-parameter co-variation while preserving idiographic first-stage estimation. For intensive longitudinal research, this creates a practical route to asking not only what is typical, but also how people differ, how those differences cohere across a dynamic system, and how person-specific dynamic signatures relate to broader substantive outcomes.

Open Practices Statement

This study was not preregistered. To support transparency, reproducibility, and reuse, the research compendium and analysis materials are publicly available on the Open Science Framework (https://osf.io/uenr7/) and GitHub (https://github.com/jeksterslab/manMetaVAR; https://jeksterslab.github.io/manMetaVAR/index.html). The research compendium associated with this manuscript corresponds to `manMetaVAR` version `0.5.8`. Source code and documentation for the `fitVARMxID` and `metaDyn` R packages are also publicly available on GitHub (https://github.com/jeksterslab/fitVARMxID; https://jeksterslab.github.io/fitVARMxID/index.html; https://github.com/jeksterslab/metaDyn; https://jeksterslab.github.io/metaDyn/index.html). In addition, to facilitate reproducibility, we provide containerized computing environments with the required software dependencies. Detailed instructions for obtaining and using these containers are available at https://jeksterslab.github.io/manMetaVAR/articles/containers.html.

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Figure 1

Overview of Monte Carlo Performance by Inferential Target and Method

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Note: Each panel summarizes a performance metric across design cells separately for population-average parameters (fixed effects) and between-person heterogeneity parameters (random effects). Points denote the mean across design cells, and intervals denote the range across design cells. Across fixed effects, the three methods were broadly similar in point-estimate performance, with BMLVAR (Bayesian multilevel vector autoregressive benchmark fit in `Mplus`) showing the best interval calibration. Across random effects, MetaVAR (proposed two-stage meta-analytic structural equation modeling approach) showed the clearest advantage, with lower absolute relative bias and RMSE and coverage closer to the nominal 95% target than either BMLVAR or Uncorr (uncertainty-uncorrected two-stage benchmark).

Figure 2

Parameter-Level Empirical Coverage of Nominal 95% Intervals Across Methods and Design Conditions

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Note: Cells show coverage relative to the nominal 0.95 benchmark for each target parameter under each simulation condition. Values near 0.95 indicate well-calibrated interval estimation, whereas lower values indicate undercoverage. The heatmap highlights two main patterns: BMLVAR (Bayesian multilevel vector autoregressive benchmark fit in `Mplus`) performed best overall for fixed-effect coverage, whereas MetaVAR (proposed two-stage meta-analytic structural equation modeling approach) showed substantially better calibration for random-effects variances and covariances. Undercoverage was most pronounced for Uncorr (uncertainty-uncorrected two-stage benchmark), especially for heterogeneity parameters and under shorter series. For compact display, parameter labels are abbreviated as follows: Sp = set point, AR = autoregressive effect, CL = cross-lagged effect, Var = variance, and Cov = covariance. Subscripts follow the model notation, with 1 and 2 denoting the two variables. T denotes the number of measurement occasions, and N denotes the number of individuals.